

Efficiency wages and unemployment

This chapter is concerned with the causes and consequences of a minimal departure from the competitive labour market model. We keep all the traditional characteristics that are usually seen as underlying competitive market behaviour such as a multiplicity of buyers and sellers, a homogeneous product, etc. We then just add the feature that the productivity of labour may depend positively on the wage. We will demonstrate that there are good justifications for this idea, and that the macroeconomic consequences, mainly with respect to structural unemployment, are important.

We start out in Section 11.1 by giving a simple example of efficiency wages. In Section 11.2 we then consider the consequences of efficiency wages in a partial equilibrium framework, focusing on the optimal wage setting in the individual firm. Section 11.3 takes one step back to discuss the reasons for efficiency wages in more detail, that is, the reasons why paying higher wages may be a way for an employer to increase the productivity of his employees. This section also considers some empirical evidence in support of efficiency wage theory. Section 11.4 analyses efficiency wages in a macroeconomic setting, developing a general equilibrium model determining the equilibrium real wage and the structural rate of unemployment. On this basis we discuss some possible structural policies to fight involuntary unemployment.

11.1 Efficiency wages: a basic example

The idea that compensation is important for productivity could well be an old one. Consider a cotton farm where labour input comes from slaves owned by the landlord. Assume that the landlord does not care about the well-being of the slaves, but sees them only as labour power. The landlord provides food and shelter for the slaves and their families and will be faced with a basic trade-off when he decides on the quantity and quality of food and shelter to provide. Think of this amount as the slaves' 'wage'.

The wage has to have a certain level just to keep the slaves alive. Above this minimum level, a higher wage will mean less miserable lives and better health for the slaves, and better health means higher productivity. So, a higher wage has a positive influence on the profits of the farm due to the improved productivity of the slaves, and a negative influence due to the higher costs of providing the slaves with the necessities of life. When the wage is still close to the minimum level, the productivity effect should be large and dominate the cost effect, because a higher wage in this range will prevent the slaves from being physically weak and ill most of the time. The wage should thus, for pure profit reasons, be raised above this minimum. However, at sufficiently high wages, the cost effect of an increased wage will dominate the productivity effect, since only a little further improvement in the slaves'

health conditions can be obtained by higher wages if the health conditions are already good. So the wage should, for pure profit reasons, be set lower than that. In between there has to be an 'optimal wage', a level of provision of food and shelter for the slaves at which the positive productivity effect and the negative cost effect just offset each other. Starting from this optimal wage, a 1 per cent increase in the wage, meaning a 1 per cent increase in the labour cost of one slave, must also imply a 1 per cent increase in the output of each slave. If the increase in productivity (production per slave) were larger than 1 per cent, then the 1 per cent increase in wage would increase profits, and the original wage could not be optimal. At the optimal wage the elasticity of productivity with respect to wage thus has to be exactly 1, a famous condition of which you will hear much more in this chapter.

Under slavery there are pure health reasons for a positive influence of wages on productivity. In modern economies health may still motivate such an influence, but as we shall see, there may be other reasons as well. In so far as there is a positive connection, a firm will be faced with a trade-off concerning wages similar to the one faced by the slave owner. This trade-off in wage determination may have serious macroeconomic implications, as this chapter will demonstrate.

A positive influence of pay on productivity can be the cause of structural unemployment and may therefore motivate structural policies intended to fight the permanent part of unemployment.

11.2 A first look at the consequences of efficiency wages in a partial equilibrium framework

A simple partial equilibrium model

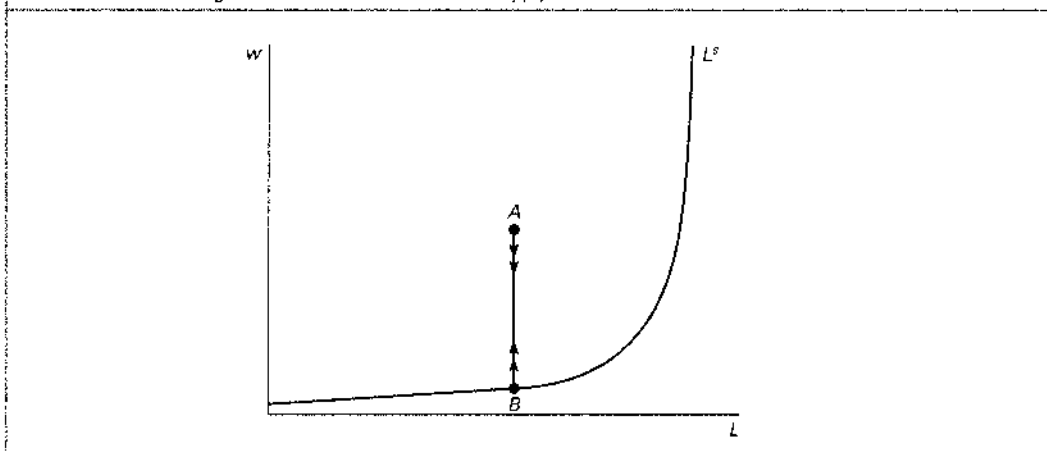
Consider a single firm that produces output from labour input L . Let the total revenue of the firm be $zR(L)$, where we assume that $R'(L) > 0$, and $R''(L) < 0$, and where z is just a multiplicative shift parameter. We can think of revenue as price times production, so $zR(L) = zp(f(L))f(L)$, where $f(L)$ is the production function, $p(y)$ is price as a function of output $y = f(L)$, and z is a (demand) shock to p , or a (supply) shock to f . If the firm is a perfect competitor on the output market the function p is constant, and in that case the revenue curve R must have its property $R'' < 0$ from $f'' < 0$. If the firm is large on the output market, $p(y)$ can be a downward-sloping curve associated with a downward-sloping (inverse) demand curve.

For a clean argument we will first, and only temporarily, depart from the conditions usually seen as underlying wage-taking behaviour and assume that the firm is a monopsonist, the sole buyer, on the labour market. The firm faces an upward-sloping labour supply curve, $L^s(w)$, coming from many wage-taking workers, as illustrated in Fig. 11.1. The firm sets the (real) wage w , and it also decides how much labour L to buy. It must, of course, obey the restriction $L \leq L^s(w)$. The profits of the firm are $zR(L) - wL$.

Such a firm would never choose a combination (w, L) above the labour supply curve, like the point A in Fig. 11.1. From a position above the labour supply curve the firm can lower the wage rate and buy the same amount of labour, that is, it can move vertically downwards from A towards B , which lowers labour costs, wL , leaves revenue, $zR(L)$, unchanged, and hence unambiguously raises profits. Likewise, moving upwards from B means buying the same amount of labour more expensively, which unambiguously lowers profits. Therefore the firm will position itself on the labour supply curve which means that there will be no unemployment: the firm wants to buy the full amount of labour supplied at the going wage rate.

This argument assumes that productivity does not depend on pay. The wage rate does not enter the revenue curve. Assume now that there is a positive influence of the wage rate on the productivity of labour, and that the relationship is given by the function $a(w)$, where a is productivity and $a' > 0$. The revenue curve will then be $zR(a(w)L)$, which we can think

FIGURE 11.1 Positioning oneself above or on the labour supply curve



of as $zp(f(a(w)L))f(a(w)L)$. A first important insight is that it is no longer *necessarily* true that moving downwards from a point like A increases profits, or that moving upwards from a point like B decreases profits. Moving upwards from B still means that the firm buys the same number of hours of work at a higher price per hour, which raises labour costs wL , but it also means that the revenue, $zR(a(w)L)$, increases, because $a(w)$ increases. The latter effect may dominate, and in this case the firm will prefer a point above B to the point B itself.

Analysing the model

The phenomenon of a positive dependence of productivity on wage is called ‘efficiency wages’ (wages that work to promote productive efficiency). We have argued that efficiency wages may undermine the argument that a monopsonist wage setter would *never* position itself above the labour supply curve. To show that the firm’s optimal combination of real wage and employment could actually be above the labour supply curve, one must study the full optimization problem of the firm which is to choose w and L to maximize profits $zR(a(w)L) - wL$, subject to the constraint $L \leq L^s(w)$. It is a usual trick to first look at the unconstrained optimization (disregarding the constraint $L \leq L^s(w)$). If this gives a combination (w^*, L^*) that fulfils $L^* \leq L^s(w^*)$, the constrained optimum will also be (w^*, L^*) .

The first-order conditions for an unconstrained maximization of $zR(a(w)L) - wL$ with respect to w and L are:

$$zR'(a(w)L)a'(w)L = L, \quad (1)$$

$$zR'(a(w)L)a(w) = w. \quad (2)$$

From the second of these, $zR'(a(w)L) = w/a(w)$. Inserting into the first condition gives:

$$\frac{a'(w)w}{a(w)} = 1. \quad (3)$$

You will recognize this as the condition we derived in the discussion of the slave economy above: the wage rate should be set at a point where the elasticity of labour productivity with respect to the real wage is 1 (if there is such a wage rate and if the optimum is interior). The intuition is also the same, but worth elaborating now that we have a formal framework.

In the absence of efficiency wages, output and revenue are produced by labour input measured in *hours*, L , and profits are $zR(L) - wL$. Whatever the optimal value of L turns out to be, it must be profit maximizing to buy that amount at the lowest possible cost, w , per unit, so the firm should set w at the lowest level required for buying L . In the presence of efficiency wages, output and revenue are produced by labour input measured in *efficiency units*, $a(w)L$, and profits are $zR(a(w)L) - wL = zR(a(w)L) - (w/a(w))a(w)L$. Whatever the optimal value of $a(w)L$ turns out to be (and note that for varying w one can keep $a(w)L$ fixed by adjusting L appropriately), it must be profit maximizing to acquire that amount of labour in efficiency units at the lowest possible cost, $w/a(w)$, per unit, so w should be set to minimize $w/a(w)$. The numerator here is the price of one hour of work, and the denominator is the number of efficiency units arising from one work hour, so the fraction is the price per efficiency unit. Hence, $w/a(w)$ should be minimized which, if the minimizing wage rate is interior (not 0 and not infinite), requires that a 1 per cent increase in w exactly gives a 1 per cent increase in a (if it gave more, $w/a(w)$ could be lowered by increasing w , etc.). This is exactly what is stated in (3), the so-called 'Solow condition'.¹

THE SOLOW CONDITION



If the labour productivity of workers in a firm depends positively on the workers' pay through an efficiency function, $a(w)$, then an (interior) optimal wage rate w^* of the firm depends only on this efficiency function, and w^* is given by the condition that the elasticity of productivity with respect to real wage is 1, condition (3) above.

To find the optimal labour demand, L^* , one inserts w^* into (2) and solves for L . For w fixed at w^* , this equation is the usual one saying that the marginal cost of one *hour* of work should equal the marginal revenue that the hour gives rise to. Still assuming an interior optimum, (w^*, L^*) is simply given as the solution to the two equations (2) and (3). It may well be that $L^* < L^s(w^*)$, and in this case the unconstrained optimum is also the constrained optimum. Furthermore, *if it happens to be the case that $L^* < L^s(w^*)$, then there is unemployment in an equilibrium with a fully adjusted wage rate.*

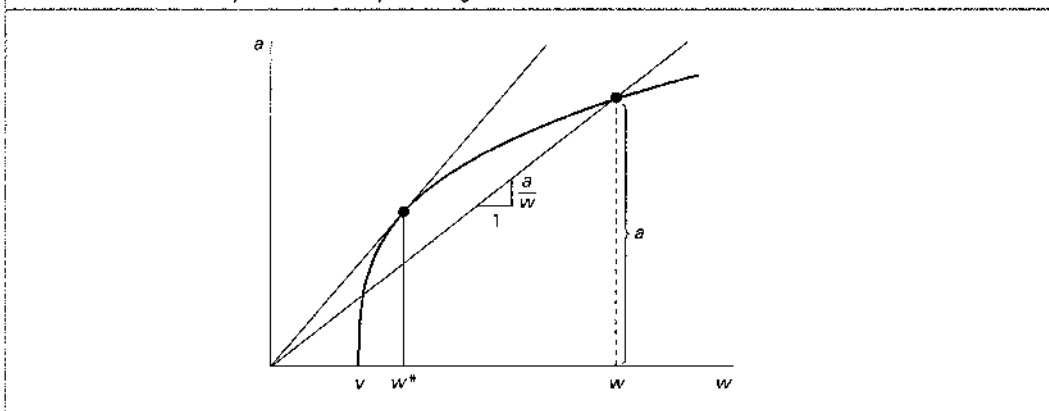
Can one be sure that the unconstrained optimum is indeed interior and that it is given by the first-order conditions? This depends on the shapes of the functions R and a . Assume that the efficiency function, $a(w)$, looks like the thick curve shown in Fig. 11.2. The wage rate w has to be above a certain minimal level, v , to obtain positive productivity at all, and the marginal increase in productivity obtained by a given increase in the wage is decreasing in w , but possibly large for wages close to v .

For any given wage rate w , the fraction $a(w)/w$ is the slope of the ray through the origin and the point $(w, a(w))$. For profit maximization this slope should be maximized, since $w/a(w)$ should be minimized. The maximal slope is obtained just where the ray becomes a tangent to the a curve as indicated by the ray most to the left in Fig. 11.2, that is, at the wage rate w^* where the slope $a'(w^*)$ of the a curve is equal to the slope $a(w^*)/w^*$ of the ray. Note that $a'(w^*) = a(w^*)/w^*$ is equivalent to the Solow condition.

Given an efficiency curve as in Fig. 11.2, the optimal wage rate is indeed interior ($0 < w^* < \infty$) and given by the Solow condition. Standard assumptions on the revenue curve, e.g. $R(0) = \infty$ and $R'(\infty) \leq 0$, ensure that the full unconstrained optimum is interior and given by the first-order conditions. The shape of the efficiency curve in Fig. 11.2 was not just chosen to ensure a well-behaved optimum. It seems reasonable, and it was the kind of shape that was implicitly argued in our slave economy example above.

1. The condition is named after the Robert M. Solow you know from growth theory. He once gave his students pretty much the profit maximization problem we study here as an exercise and asked them to derive the Solow condition.

FIGURE 11.2 An efficiency curve and the optimal wage rate



We started this chapter by saying that we would consider a minimal departure from the competitive labour market and keep all the other characteristics usually seen as underlying wage-taking behaviour. We then, nevertheless, assumed that the firm was a monopsonist on the labour market. However, if there were many buying firms the labour market would, with the efficiency wage effect, work essentially as under monopsony.

Usually we would say that many firms should imply that each firm is a wage taker, so no firm sets a wage rate. The informal argument behind this is that if there are many buyers, *no buyer can get away with lowering the price* (compared to the 'going market price'), because then no one would want to sell to that buyer, and *no buyer would be interested in setting a higher price*, since the buyer already gets the demanded quantity at the going price. Therefore each buyer takes the price as given. Here, however, we have a situation where a buyer may want to pay a price above the market clearing level. Nothing prevents a buyer from setting a higher price. If there were many firms on the labour market, each identical to the firm considered above, then each of these firms would have an unconstrained optimum, (w^*, L^*) , identical to the one described above, and total (unconstrained) labour demand would be L^* times the number of firms which could still be smaller than $L^*(w^*)$. There would then be unemployment in an equilibrium with fully adjusted wages. Thus we do not have to think of the labour market as a monopsony. All the characteristics usually seen as underlying perfect competition, many buyers and many sellers, etc., can be fulfilled, and still, with the efficiency wage effect, the labour market will work as described.

Discussion of the model's main implications

The partial equilibrium model above shows that efficiency wages can *potentially* cause unemployment in an equilibrium with fully adjusted wages if $L^*(w^*)$ happens to be greater than L^* times the number of firms. In the partial equilibrium model efficiency wages do not *necessarily* imply unemployment.

Assuming that there is unemployment in equilibrium, the model has the implication that variations in the supply/demand shift parameter z do not affect w , since the optimal wage rate is given entirely by the efficiency function $a(w)$. Changes in z only affect L , and high values of z go with high values of L .

This fits well with, and can be part of an explanation of, one of the most important empirical facts of the business cycle: that real wages are weakly correlated with output over the cycle, while employment varies positively and is strongly correlated with output. For this consideration we view z as a shock variable.

However, the same result does not fit well with the stylized fact of growth, that in the long run real wages increase in proportion to labour productivity. Now we view z as a

long-run trend in productivity. In the model above, a steady and gradual increase in z will have no influence on the real wage rate. Counterfactually, the real wage will remain at a given level despite the steady growth in productivity.

One might think that this is a serious problem for the efficiency wage model. However, the phenomenon is an artefact of the simple partial equilibrium framework we have considered. We will argue below that the productivity of the workers in each firm should really depend on the excess of the wage paid in the firm over the 'normal income' in society. For instance, what makes workers provide more effort is not the wage as such, but a wage rate in excess of what could normally be expected. Furthermore, if there is a trendwise increase in normal incomes due to an increasing trend in productivity, it should be the *relative* excess of the wage over normal income that motivates effort. So the efficiency function should be like $a(w) = \bar{a}([w - v]/v)$, where $\bar{a}$ is an increasing function and v is 'normal income'. In that case the optimal wage in each firm will be determined relative to v , and if v increases, so will w , while as long as v stays relatively constant, so will w . Hence, if the expected normal income is not affected (much) by short-run shocks to z , but is affected by permanent trend changes in z , then we can have a situation where the real wage rate does not respond (much) to random fluctuations in productivity, but does increase proportionally to permanent productivity. Exercise 2 as well as Exercise 6 below will ask you to perform some further analysis highlighting this point.

The essence of the explanation of unemployment offered by efficiency wage theory is this: employers (not employees) drive up wages to attain an optimal level of productivity from each hired worker, and the wages thus determined (by the employers) may be so high that the employers do not want to hire all the workers who want jobs at those wages. The unemployment that results is truly involuntary. The workers really want to have jobs at the going wages, and they are in no way responsible for wages being so high.

UNEMPLOYMENT AND FLUCTUATIONS IN THE PARTIAL EQUILIBRIUM EFFICIENCY WAGE MODEL



A positive dependence of workers' productivity on their pay may imply that firms, in order to extract an optimal work effort from each worker, drive up wages to a level where employers do not want to hire all the workers who would like to work. Hence efficiency wages can help to explain structural unemployment, that is, the coexistence of some unemployment and fully adjusted prices and wages. In the partial equilibrium efficiency wage model with unemployment, temporary demand and supply shocks do not affect the real wage, but give rise to fluctuations in employment.

The plausibility of the efficiency wage explanation of structural unemployment stands and falls with the plausibility of the efficiency wage phenomenon. Are there good reasons to believe that productivity is positively affected by wages?

11.3 The causes of efficiency wages

The four traditional reasons

Besides the health argument, which may not be very important in developed economies, usually four reasons are given for why labour productivity depends positively on (above average) pay.²

2. For more about these reasons and their implications, consult the short and informative survey by Janet Yellen, 'Efficiency Wage Models of Unemployment', *American Economic Review*, 74, 1984, pp. 200–205.

1. Keeping the workers in the firm

Paying wages that are high relative to what the workers could otherwise earn gives the workers an incentive not to quit their jobs so easily. This reduces labour turnover in the firm and saves the firm costs of recruiting new employees who are not fully productive until they have been trained. Thus, on average the productivity of the workers in the firm will be higher.

2. Getting good workers when hiring

Every now and then some workers quit their jobs for various reasons and the firm will want to hire new employees. It is realistic to assume that when it hires new workers, the firm cannot perfectly observe the abilities of each applicant, and that workers with higher ability will also demand a higher wage to be willing to accept a job offer. By offering relatively high wages, the firm will thus be able to recruit the more able applicants, thereby obtaining a more favourable composition of its workforce. This raises average labour productivity in the firm.

3. Making the workers in the firm work hard – 1

Assume realistically that the firm cannot perfectly monitor the effort exerted by each and every worker all the time, but that it can from time to time, by some control mechanism, observe the effort levels of some of the workers. Assume further that workers who are caught not exerting an effort level as required by the firm are fired. A relatively high wage will then create an incentive for the individual worker to work hard (as hard as required) since the cost of being fired is the difference between the wage paid in the firm and the income that could be earned outside the firm through unemployment benefits or alternative jobs. Working less hard than required by the employer is sometimes called ‘shirking’, but note that the effort level required by the employer could well be high. The model of efficiency wages based on the motive explained here is often called the ‘shirking model’.

4. Making the workers in the firm work hard – 2

While the incentive effect just described builds fully on self-interested ‘rationality’ in the usual sense, it may also be that workers who are paid well compared to what is normal will work hard simply because they feel that they are treated well and want to be nice to people who are nice to them. Note that here it is not the risk of being fired that motivates the worker to work hard, but pure *reciprocity*: the wish to respond to nice behaviour with nice behaviour. This is not irrationality. What is argued is that ‘treating others as they treat you’ increases a person’s well-being, and therefore reciprocity is in full accordance with utility maximization. It is not just individual income and consumption that give utility. The other side of reciprocity is that a low wage will be responded to by low effort.³

A closer look at the shirking model

We will present the shirking model of efficiency wages in a little more detail to illustrate how the positive effect of wages on productivity may be given a microeconomic foundation.⁴ Consider a firm that is currently employing a certain number of workers. Assume that for each worker the utility of working in the firm at the wage rate w exerting the effort level a is

3. The reciprocity-based model of efficiency wages was presented in the article by George A. Akerlof and Janet L. Yellen, ‘The Fair Wage-Effort Hypothesis and Unemployment’, *Quarterly Journal of Economics*, **105**, 1990, pp. 255–283. The article reports plenty of evidence that high wages in a firm relative to ‘normal wages’ imply high effort in the firm in question, and that low wages tend to create low effort, for ‘emotional’ reasons based on feelings of loyalty, gratitude, anger, injustice, etc.

4. In this subsection we study some elements (the worker incentive part) of the theory suggested by Carl Shapiro and Joseph E. Stiglitz in ‘Equilibrium Unemployment as a Worker Discipline Device’, *American Economic Review*, **74**, 1984, pp. 433–444.

$w - c(a)$, where $c(a)$ is a convex effort cost function, $c' > 0$, $c'' > 0$, $c'(0) = 0$, and $c'(a) \rightarrow \infty$ as $a \rightarrow \bar{a} > 0$, where possibly $\bar{a} = \infty$. The worker's utility outside the firm would be v , which reflects what can be obtained by unemployment benefits, home production, alternative jobs, etc. The worker gets utility v if he or she is fired.

The employer presents the worker with a contract $(w, \hat{a})$ that specifies the wage w paid and the effort level $\hat{a}$ demanded, where $\hat{a} < \bar{a}$. The employer can only imperfectly monitor the worker's effort. Shirking, which means $a < \hat{a}$, is discovered with probability q , where $0 < q < 1$. A worker who is discovered shirking will be fired. The expected utility of an employed worker in case $a \geq \hat{a}$, which is non-shirking, is $w - c(a)$, and in case $a < \hat{a}$, which is shirking, the expected utility is $(1 - q)[w - c(a)] + qv$.

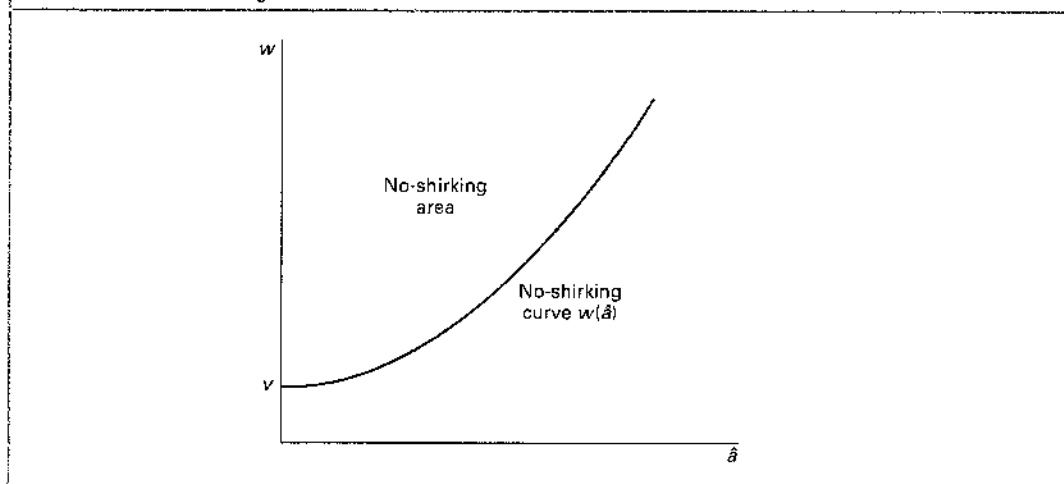
Since $c(a)$ enters negatively in the worker's utility, it never pays to provide 'too much' effort, so if a worker does not shirk the effort will be exactly as required, $a = \hat{a}$, and if a worker does shirk the effort will be $a = 0$. Therefore the condition stating that non-shirking is at least as good as shirking is $w - c(\hat{a}) \geq (1 - q)w + qv$, or:

$$w \geq v + \frac{c(\hat{a})}{q}. \quad (4)$$

This is the *no-shirking condition*, and it is illustrated in Fig. 11.3. From the point of view of the employer this condition states how high a wage must be paid for its workers to provide a given demanded effort level $\hat{a}$. Note that the required wage is increasing in the demanded effort $\hat{a}$, decreasing in the probability q of detecting shirking, and larger than, and increasing in, the alternative income v .

Since it does not pay for the employer to offer a higher wage than required for a given effort, one may as well write the no-shirking condition with equality. This gives the combinations of $\hat{a}$ and w indicated by the curve in Fig. 11.3. One can read the condition inversely as stating how large an effort each worker will provide for a given wage rate w . This corresponds to 'reading' Fig. 11.3 from the w -axis, that is, with w as the independent variable, and from this point of view the curve in Fig. 11.3 looks exactly like the curve in Fig. 11.2 above. The curve in Fig. 11.3 emerges from solving the no-shirking equation for $\hat{a}$. With the assumed properties of the function $c(a)$, this can always be done. One will then arrive at a function giving the effort level as a function of w , v , and q , that is, a function $\hat{a}(w, v, q)$. This function will only assume positive values for w larger than v , it will be increasing in w , decreasing in v , and increasing in q .

FIGURE 11.3 The no-shirking condition



One particular form of the effort cost function that fulfils all the assumptions above and allows us to solve the no-shirking equation analytically is $c(a) = \beta a^{1/\eta}$, where $\beta > 0$ and $0 < \eta < 1$. Using (4) with equality gives:

$$a = \hat{a}(w, v, q) = \left[\frac{q}{\beta}(w - v) \right]^\eta. \quad (5)$$

The efficiency function that was considered in the partial equilibrium model of Section 11.2 could well have taken the form $a(w) = k(w - v)^\eta$, where we have defined $k \equiv (q/\beta)^\eta$. Indeed, in Fig. 11.2 the efficiency function was drawn exactly that way.

The shirking model of efficiency wages studied in this section gives mainly three insights. First, it gives us an optimizing behaviour rationale for the idea of a positive effect of wages on productivity. Second, it suggests that it is only for wages above a certain base level that the positive effect on productivity starts to be felt. Third, it gives the interpretation of that base level as, more or less, the 'normal' income that a worker could obtain if separated from the firm. This normal income is therefore often referred to as the worker's 'outside option'.

Note that the other three motives for efficiency wages mentioned above have the same character. It is only if a firm's wage is above normal pay that workers will have a special incentive to stay in a firm, or to apply for a job in a firm, or have reason to feel particularly well treated by the firm. Therefore it is only for wages above an alternative or normal level that one should expect positive effort, and more effort for higher wages. So, all the motivations for efficiency wages point to efficiency functions like $a = k(w - v)^\eta$.

Empirical evidence in support of efficiency wages

We have already mentioned in Footnote 3 that Akerlof and Yellen have provided empirical evidence supporting their theory of efficiency wages based on reciprocity. A survey by the Swedish economists Jonas Agell and Helge Bennismarker is worth mentioning as well.⁵ The paper presents the results from a survey among Swedish human resource managers. The survey is more representative than earlier surveys and includes many firms (1,200) and covers the relevant production sectors and firm size categories. The response rate obtained was very high (75.1 per cent).

An additional feature that makes the Swedish survey particularly interesting is that by the spring of 1999, when the survey was done, unemployment was high, around 10 per cent after having peaked at 13.6 per cent in 1994, while inflation was close to zero and had been very low for some years, on average 1 per cent per year over the preceding five years. This is a good environment for studying wage rigidity and resistance to wage reductions. Since unemployment was high, real wages should fall in the absence of real wage rigidity, and since inflation was low, real wage cuts of substantial size could only be achieved by nominal wage cuts. Hence, to see if firms resisted real wage cuts one could simply ask the human resource managers if they had cut (nominal) wages and related questions.

Table 11.1 presents some of Agell and Bennismarker's results in the form of some of the questions asked and some crude statistics on the answers obtained. From the table's first line it appears that only 3.2 per cent of the firms answered in the affirmative to the question whether they had cut regular base pay at any time during the crisis years of the 1990s (and most of the firms that had cut wages reported that they had done so for only a few employees, not for employees in general). This means that wage cutting by firms was not widespread despite very high unemployment.

5. Jonas Agell and Helge Bennismarker, 'Wage incentives and Wage Rigidity: A Representative View from Within', *Labour Economics*, 14, 2007, pp. 347–369.

TABLE 11.1 Selected results from Swedish survey by Agell and Bennmarker

Question	Answer
1. Cut base pay during crisis years of the 1990s?	3.2% affirmative
2. Encountered undercutters?	13.5% affirmative
3. Rejected offers by undercutters?	89.6% affirmative
4. Undercutting rejected because it 'violates firm's personnel policy; creates internal conflict'?	29.5–41.7% affirmative depending on sector
5. Extent to which it can be evaluated if an employee performs satisfactorily?	49.3% less than certain about performance
6. Implication for work effort if wages/salaries are increased in comparable firms?	65.9% answered impairment, great or small
7. Do employees who are dissatisfied with their pay normally reduce effort?	49.0% affirmative
8. Do employees who are dissatisfied with their pay normally seek employment elsewhere?	58.6% affirmative

Source: Jonas Agell and Helge B  n  marker, 'Wage Incentives and Wage Rigidity: A Representative View from Within', *Labour Economics* 14 (2007), pp. 347–369.

The next two lines are related to the first one. Active underbidding by workers (job seekers offering to work under conditions inferior to those normally offered to new employees) was quite rare despite the circumstances. Even more striking, when the firms were faced with undercutters a huge 89.6 per cent of them rejected the offers, which may explain why undercutting is so rare.

These findings are consistent with the theory of efficiency wages. As we have just seen, to induce an appropriate level of productivity each firm must pay an *individual* wage that is sufficiently high relative to *normal* pay, or the outside option. This outside option is, of course, heavily influenced by what other firms pay. Hence, the individual firm may be reluctant to be the first to cut wages because, as long as the other firms have not yet cut their wages, this will imply a substantial loss of productivity. If the firms could coordinate to cut wages simultaneously the productivity drop would not be large because normal pay would drop at the same time, but such a coordination task is too complex in an economy with hundreds of thousands of firms. Hence the theory of efficiency wages may imply some 'relative wage resistance' (reluctance to decrease one's own wage compared with the wages of other firms) that can provide an explanation for the findings in the first three lines of Table 11.1.

Other explanations could be possible, and therefore the survey of Agell and B  n  marker includes some questions that relate more directly to the theory of efficiency wages. As one can see from the fourth line of the table, when the managers were asked to explain *why* they would not accept the offers of undercutters, a substantial fraction of them indicated the reason that acceptance would 'violate the firm's personnel policy; create internal conflict', a motivation clearly related to how pay affects average productivity.

The managers were also asked how well they can monitor work effort. This is of importance for the shirking model where the assumption of imperfect monitoring is essential. The fifth line of the table shows that around half the firms indicated that they were less than certain about work performance.

The table's sixth line shows the question of most direct relevance for efficiency wages. It asks the subtle question how workers' effort would be affected by a wage increase in *other* firms. According to efficiency wage theory this should reduce effort by increasing the workers' outside option. As can be seen, a great majority of firms answered that a wage increase in other firms would indeed affect work effort in their own firm negatively.

The seventh and eighth lines of the table show that it is widely believed among managers that employees who feel their pay is too low will reduce their effort or seek employment elsewhere. These answers are in line with the reciprocity motivation and the labour turnover motivation for efficiency wages, respectively.

Overall the survey by Agell and Bennismarker provides a good deal of evidence that the productivity of a firm's employees is positively affected by the excess of the firm's wages over normal incomes.

FOUNDATIONS OF EFFICIENCY WAGES



Theory and evidence suggest that workers' productivity depends positively on their pay. From a theoretical as well as an empirical perspective, it is the *excess of the firm's wages over normal incomes* that should affect the productivity of employees positively. The positive dependence of productivity on above-normal pay is the central idea of efficiency wage theory.

11.4 Efficiency wages in a macroeconomic framework

In Section 11.2 we took the revenue curve of the individual firm as given. This was what made it a *partial equilibrium* analysis. We will now consider the full production side of the economy and be specific about how each firm's revenue is associated with the demand for its output, and how this demand is derived from the behaviour of the household sector of the economy. Our analysis will consequently be one of *general equilibrium*. This will enable us to show that the equilibrium unemployment rate in an economy with efficiency wages will indeed be positive; a result we could not establish with certainty in our partial equilibrium framework.

We will study a general equilibrium model intended to picture relevant aspects of the entire (macro)economic system, just as we did in Parts 1, 2 and 3 on growth theory. There will be an important difference, however. The model of this chapter will have only one period and hence no capital and no savings. The only input in production will be labour, and the only task of consumers will be to allocate their total income and consumption across different products. This is because our focus is not on growth and capital accumulation, but on the determination of the rate of structural unemployment in a single period. As usual in economics we simplify dramatically, so our model will involve only the few aspects we find most important for the problem we are analysing.

A macroeconomic framework

We will take the view that the product of each firm is a little different from the product of any other firm, so that each firm is the sole producer of a specific type of output. Hence each firm is a 'local monopolist' and sets the price of the commodity type it produces. This does not mean that each firm necessarily has a lot of market power. Its output may be in close substitution with the products of other firms, making demand highly sensitive to price, implying that the control over the price cannot be exploited much.

Since firms are not price takers, but set prices themselves, there is imperfect competition in product markets. The particular form of imperfect competition where each firm is a local monopolist is called *monopolistic competition*. To be specific, we assume that there are n firms each producing a separate type of output. The number of firms is large so *each firm is large in its own output market, but small relative to the entire economy*.

The demand (curve) for each output type $i = 1, \dots, n$ is

$$D(P_i) = \left(\frac{P_i}{P} \right)^{-\sigma} \frac{Y}{n}, \quad \sigma > 1, \quad (6)$$

where P_i is the money price of output type i , and P is a price index for, or an average of, the money prices P_j of all output types $j = 1, \dots, n$, so $P = P(P_1, \dots, P_n)$, and $P(x, \dots, x) = x$ for all x . The variable Y is an indicator of total demand in the economy and taken as given by the individual firm, since each firm is small relative to the economy. Equation (6) says that total demand is distributed over the output types in a way that reflects their relative prices: each product type gets the fraction $(1/n)(P_i/P)^{-\sigma}$ of total demand. There is a specific formula for $P(P_1, \dots, P_n)$ that is appropriate given that the demand curves in (6) are supposed to come from households optimizing with specific utility functions, but we do not need the exact formula for our present purposes.

Since all the n firms are faced with identical demand curves and will be assumed to have identical production functions, the optimal prices they set *in equilibrium* (to be derived below) will be identical, and hence the value of the price index will be equal to this common price. In equilibrium therefore $P_i/P = 1$ for all i , and hence in equilibrium the demand for each firm's output is Y/n . Since each firm satisfies all demand, Y/n will also be the production of each firm, and thus the total production, or the GDP, will be n times Y/n , that is, Y . Hence the indicator of total demand, Y , is the economy's GDP and it will be determined endogenously in our complete theory.

According to (6), the demand for output of type i does not only depend on P_i , as perhaps suggested by the notation $D(P_i)$, but on all of P_i , P , Y and n . However, the individual producer of output type i only has influence over P_i . The demand curve $D(P_i)$ is *iso-elastic* and its constant price elasticity is σ , that is, $-D'_i(P_i)P_i/D_i(P_i) = \sigma$ (show this by taking logs on both sides of (6) and differentiate with respect to P_i). Note that we assume $\sigma > 1$. Otherwise (if $0 < \sigma < 1$) our theory would not be meaningful.⁶

This description of the economy's demand side is compatible with utility maximization by a household sector with a particular type of utility function, a so-called constant elasticity of substitution, CES, utility function, where the (constant) elasticity of substitution is σ . For our purposes we do not need the exact derivation, but Exercise 3 will ask you to do it yourself. As a by-product you will also get the appropriate formula for the price index P .

From (6) it is clear that σ measures the degree of substitution between the commodities. The larger σ is, the more sensitive is the demand for output of type i to P_i/P , since a higher σ means that consumers are more willing to substitute away from any specific output type towards other types if its relative price increases. Therefore the market power of each firm is smaller the larger σ is. In the limit where σ goes to infinity the firm has no market power at all, since if it tries to set P_i above P , it will lose all demand. So, $\sigma \rightarrow \infty$ corresponds to price-taking competitive behaviour which is thus contained as a special case in our analysis.

The production function of each firm i is simply:

$$Y_i = a_i L_i, \quad (7)$$

where L_i is labour input and a_i is the productivity of labour input. All firms are assumed to use one and the same type of labour, and this comes from the household sector of the economy. In the labour market each firm is just one out of many buyers, and labour market conditions are similar to those underlying perfect competition (many buyers and sellers, homogeneous product, etc.). An efficiency wage effect can, nevertheless, make each firm a

6. From microeconomics it is well known that a monopolist faced with a demand curve with an elasticity that is smaller than 1 everywhere should let production go to 0 and sell its infinitesimal production at prices going to infinity. Explain why this is the case.

wage setter, as we explained in Section 11.2. In accordance with the idea of efficiency wages, the labour productivity, a_i , in firm i is assumed to depend on the real wage, $w_i \equiv W_i/P$, paid by firm i , and on the 'normal' real income, v , that a worker can obtain outside firm i :

$$a_i = a(w_i) = (w_i - v)^\eta, \quad 0 < \eta < 1. \quad (8)$$

Compared to (5) we have normalized so that $k \equiv (q/\beta)^\eta = 1$. Again, a_i really depends on both w_i and v , but the firm only controls w_i , and has negligible influence over v , since the 'normal income' relates to the whole economy. This justifies using $a(w_i)$.

Price and wage setting by the individual firm

In Section 11.1 we assumed that the firm decided on wage and employment. We could proceed similarly here and derive the revenue curve of each firm from (6) and (7), express profits in terms of W_i and L_i , and find the optimal values for W_i and L_i . This time, however, it is most convenient to proceed by determining directly the optimal values for W_i and P_i . Of course, given these (and P and v), an optimal labour demand, L_i , follows.

The real profit of firm i is $\Pi_i = (P_i Y_i - W_i L_i)/P = Y_i [P_i/P - (W_i/P)/(Y_i/L_i)]$ or, inserting for Y_i from the demand function, and $Y_i/L_i = (w_i - v)^\eta$ from the production and efficiency function:

$$\Pi_i = \left(\frac{P_i}{P}\right)^{-\sigma} \frac{Y}{n} \left(\frac{P_i}{P} - \frac{W_i/P}{(W_i/P - v)^\eta}\right). \quad (9)$$

This may be written in terms of the relative, or real, prices $p_i \equiv P_i/P$ and $w_i \equiv W_i/P$:

$$\Pi_i = p_i^{-\sigma} \frac{Y}{n} \left(p_i - \frac{w_i}{(w_i - v)^\eta}\right). \quad (10)$$

The task of the firm is to choose p_i and w_i to maximize real profits. (The firm really chooses the money price, P_i , and the money wage, W_i , such that at the given P , the desired p_i and w_i are established.) The first-order conditions for this optimization are necessary and sufficient.

Let us first take w_i as given – consider it as already chosen optimally – and look for the optimal p_i . This is a standard monopoly problem with iso-elastic demand and constant marginal cost. It is well known from microeconomic theory that the firm sets its price as a mark-up over marginal cost, where the mark-up factor is $m = \sigma/(\sigma - 1)$. To confirm, consider the first-order condition for maximizing Π_i with respect to p_i :

$$\frac{\partial \Pi_i}{\partial p_i} = -\sigma p_i^{-\sigma-1} \frac{Y}{n} \left(p_i - \frac{w_i}{(w_i - v)^\eta}\right) + p_i^{-\sigma} \frac{Y}{n} = p_i^{-\sigma-1} \frac{Y}{n} \left[-\sigma \left(p_i - \frac{w_i}{(w_i - v)^\eta}\right) + p_i\right] = 0.$$

Here the square bracket must be 0. This gives $p_i(\sigma - 1) = \sigma w_i/(w_i - v)^\eta$, or:

$$p_i = \frac{\sigma}{\sigma - 1} \frac{w_i}{(w_i - v)^\eta} = m \frac{w_i}{(w_i - v)^\eta}. \quad (11)$$

Note that unit labour cost is w_i/a_i , the cost w_i per hour of work divided by productivity a_i , where $a_i = (w_i - v)^\eta$.

Next consider p_i as already set (optimally) and look for the optimal w_i . The first-order condition is:

$$\frac{\partial \Pi_i}{\partial w_i} = -p_i^{-\sigma} \frac{Y}{n} [(w_i - v)^{-\eta} - \eta w_i (w_i - v)^{-\eta-1}] = -p_i^{-\sigma} \frac{Y}{n} (w_i - v)^{-\eta-1} [w_i - v - \eta w_i] = 0.$$

The square bracket must be 0, which is equivalent to:

$$w_i = \frac{v}{1 - \eta}. \quad (12)$$

The optimal real wage is a mark-up over the alternative or normal real income v , and the mark-up factor, $1/(1 - \eta)$, is larger the larger the productivity effect of higher wages.⁷

In view of Section 11.2, the expression for the optimal real wage in (12) should be equivalent to the Solow condition in the present setting. From (8) one gets for the elasticity of the efficiency function, $\alpha'(w_i)w_i/\alpha(w_i) = \eta w_i/(w_i - v)$. Equalizing this to 1 gives exactly (12). The intuition for this formula is thus as explained in Sections 1 and 2.

The wage curve

The normal real income v should measure what a worker in firm i could expect to earn in the rest of the economy if separated from firm i . This must depend on the risk of becoming unemployed and the income obtained in that case, and on the chance of getting employed and the real wage that can be expected in that case. We will focus on the simplest possible formulation where v is exactly the expected income of a person who becomes unemployed with probability u , the rate of unemployment in the economy, in that case receiving a real unemployment benefit b , and becomes employed with probability $1 - u$, in that case earning an expected real wage of w :

$$v = ub + (1 - u)w. \quad (13)$$

The rate of unemployment benefit b is (for now) assumed to be a fixed amount in real terms and, of course, measured in the same units as the real wage rate w , so if w is pay per day (or hour), so is b . We can interpret b to include the value of an unemployed person's home production.

The 'outside option', v , is the same from the point of view of all firms i . According to (12) all firms therefore set the same real wage rate which must then be equal to w , the expected real wage of an employed worker. Writing w for the w_i on the left-hand side of (12), and inserting the expression in (13) for v on the right-hand side gives:

$$w = \frac{ub + (1 - u)w}{1 - \eta},$$

which can be solved for w :

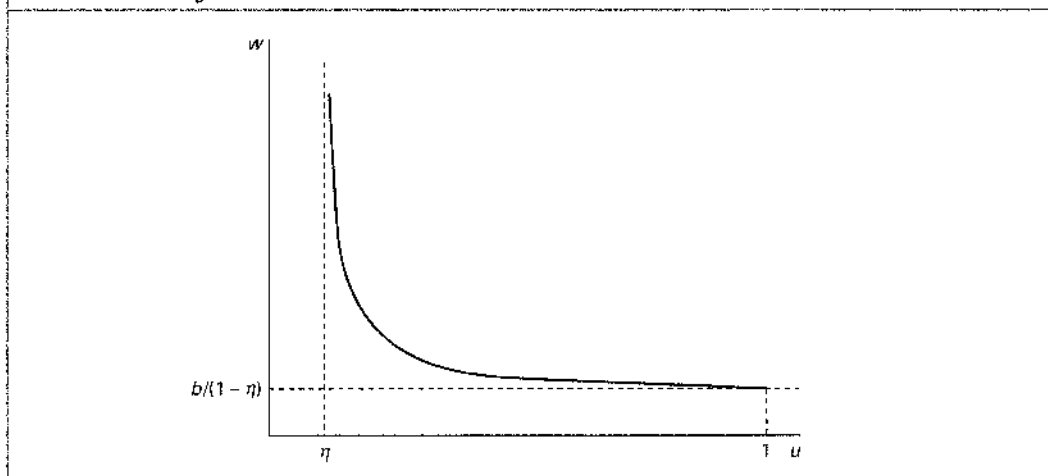
$$w = \frac{u}{u - \eta}b = \frac{1}{1 - \frac{\eta}{u}}b. \quad (14)$$

The real wage is 'marked up' over the real rate of unemployment benefit with a mark-up factor of $u/(u - \eta) > 1$, so $w > b$.⁸ Given b , the formula in (14) gives the real wage w as a *decreasing* function of the rate of unemployment, as illustrated in Fig. 11.4. Since $w > b$, higher unemployment means a lower value of the outside option in (13) and therefore the firms can obtain the same level of worker productivity for lower wages. In other words, if unemployment increases, wage pressure decreases. This is the essential incentive effect captured by the relationship in (14), a relationship called the 'wage-setting curve' or just the 'wage curve'.

7. As an exercise, show that the second-order conditions $\partial^2 \Pi_i / \partial^2 p_i < 0$ and $\partial^2 \Pi_i / \partial^2 w_i < 0$ are fulfilled at the p_i and w_i we have found.

8. For this to be meaningful one should have $u > \eta$, which we assume. It is demonstrated below that indeed in a full equilibrium of the model this condition is satisfied.

FIGURE 11.4 The wage curve



THE WAGE CURVE



The general equilibrium efficiency wage model gives rise to a negative relationship between overall unemployment and real wages, a so-called wage curve: a lower rate of unemployment means that the alternative or normal income, the outside option, as viewed from each firm will be more favourable for workers. Firms will therefore have to pay higher wages to obtain the same level of effort and this creates increased wage pressure.

The economists David G. Blanchflower and Andrew L. Oswald have written a book entirely on the wage curve.⁹ Much of the book is concerned with empirical tests and estimates. They find quite robustly (with varying methods of estimation and for several countries) that a wage curve exists, and estimates of the elasticity of the real wage with respect to the rate of unemployment are around -0.1 , meaning that a 10 per cent increase in the rate of unemployment, e.g. from 5 to 5.5 per cent, should lead to a 1 per cent decrease in real wages. This magnitude also has an implication for how large one should expect η , the real wage elasticity of effort or productivity, to be. You will be asked to think about that in an exercise.

In the following it will be convenient to express the wage curve in terms of the rate of employment $e = 1 - u$, such that w is an increasing function of e :

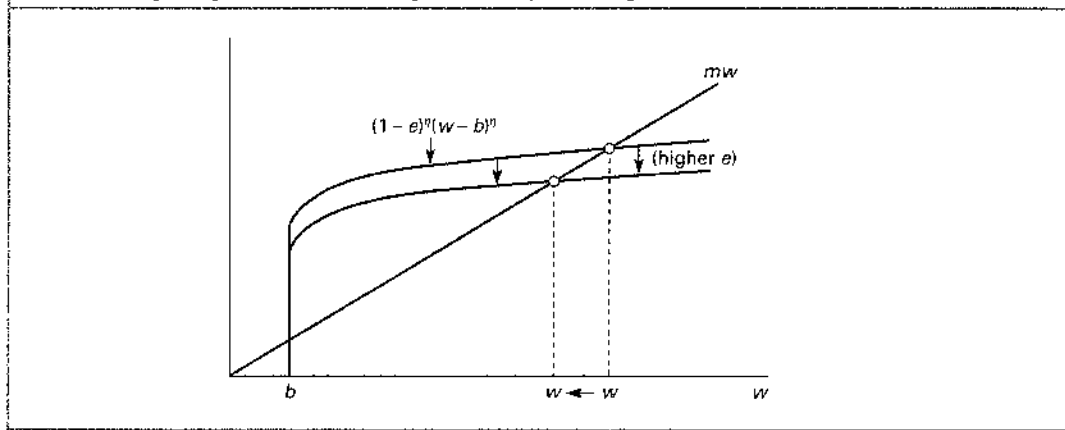
$$w = \frac{1 - e}{1 - e - \eta} b = \frac{1}{1 - \frac{\eta}{1 - e}} b. \quad (15)$$

The wage curve in this form is illustrated as the curve labelled WS in Fig. 11.6 on page 318. Note that the wage curve will shift upwards if b or η increases (try to give the intuitive explanations yourself).

The price curve

Since all firms set the same wage rate w , it follows from (11) that they also set the same relative price $p_i = P_i/P$. Since P is given for each of them, this implies that they all set the

9. David G. Blanchflower and Andrew L. Oswald, *The Wage Curve*, MIT Press, 1996.

FIGURE 11.5 Higher e gives lower w according to the firms' price setting

same nominal price P_i , and hence the value of the price index P , an average of the individual nominal prices, must be equal to this common nominal price. Hence $p_i = P_i/P = 1$ for all firms i . It then follows from (11) that:

$$1 = m \frac{w}{(w - v)^\eta} \text{ or } mw = (w - v)^\eta.$$

Inserting the value of the normal income v from (13) gives:

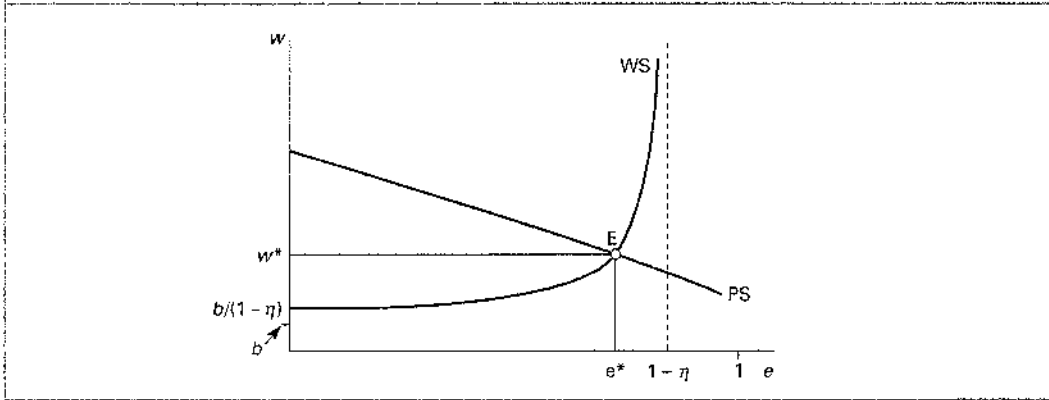
$$mw = [u(w - b)]^\eta = (1 - e)^\eta (w - b)^\eta. \quad (16)$$

A real wage w must fulfil this equation in order to be compatible with the mark-up pricing behaviour of firms. The equation does not give a closed form solution for w , but rather gives w as the intersection between the left-hand side, the straight line mw , and the right-hand side, the curved function $(1 - e)^\eta (w - b)^\eta$ of w (see Fig. 11.5). For b not too large there will, for all sufficiently small e , be two intersections. However, it is only the upper one indicated by a small circle in Fig. 11.5 that corresponds to an optimum, because one can show from (15) and (16) that at an optimum the slope of the curved function must be smaller than the slope, m , of the line.¹⁰

The fact that the upper intersection is the right one means that an increase in e implies a decrease in w , since a higher e shifts the concave curve downwards (see Fig. 11.5 again). This is because a higher rate of employment increases the value of the outside option v (remember $w > b$), which decreases productivity so that unit labour cost, $w/(w - v)^\eta$, increases, pushing up real prices in accordance with (11), and hence reducing real wages. This is the basic incentive effect underlying the 'price curve' or 'price-setting curve', giving the real wage rate w as a decreasing function of the rate of employment e . The curve is illustrated in Fig. 11.6 and labelled PS. As explained, the price-setting curve is decreasing because of the way a change in the rate of employment (or unemployment) affects worker productivity through the value of the normal income, v . If one disregards that effect by assuming $\eta = 0$, then the price curve becomes flat: (16) with $\eta = 0$ gives $mw = 1$ or $w = 1/m$, simply expressing mark-up pricing by the firms, $P = mW$, when there is a constant labour productivity of unity.

A higher unemployment benefit, b , as well as more market power in product markets, a higher m , shifts the price-setting curve downwards, since both will shift the concave curve in Fig. 11.5 downwards for given e (give the intuitive explanations yourself).

10. The slope of the curved function is $(1 - e)^\eta \eta (w - b)^{\eta-1} = \eta (1 - e)^\eta / (w - b) (w - b)^\eta$, which, using (16), is equal to $\eta mw / (w - b)$. Inserting the optimal value for w from (15) gives the slope $(1 - e)m$, which is smaller than m as long as $e > 0$.

FIGURE 11.6 Labour market and macroeconomic equilibrium

Macroeconomic equilibrium

Figure 11.6 shows both the wage curve and the price-setting curve in the same diagram. The diagram assumes that the value of the benefit rate b is sufficiently low to ensure that the decreasing price-setting curve PS reaches far enough to the south-east to ensure an intersection with the wage curve WS. The full labour market equilibrium is at the point E, where the two curves intersect each other. Only the employment–wage combination (e^*, w^*) of the intersection is compatible with both the wage-setting and the price-setting decisions of the firms. In other words, e^* and w^* are the solutions in e and w to the two equations (15) and (16), repeated here for convenience:

$$w = \frac{1-e}{1-e-\eta}b, \quad (\text{WS})$$

$$mw = (1-e)^\eta(w-b)^\eta. \quad (\text{PS})$$

Thus in macroeconomic equilibrium e^* and w^* depend (only) on the parameters η , σ (through m), and b . Note from Fig. 11.6 that $e^* < 1 - \eta$, which is equivalent to $u^* = (1 - e^*) > \eta$. The condition mentioned in Footnote 8 is indeed fulfilled in equilibrium.

However, the labour market equilibrium illustrated in Fig. 11.6 also determines the GDP of the economy. Given a total (inelastic) labour supply, $\bar{L}$, total employment will be $e^*\bar{L}$. In equilibrium this is distributed evenly across the n firms, so employment in each firm will be $e^*\bar{L}/n$. Since w^* has also been determined, the productivity in each firm will be given by this w^* and the v implied by w^* and $u^* = 1 - e^*$, that is, $a^* = [(1 - e^*)(w^* - b)]^\eta$. Hence the GDP of this economy is $Y = [(1 - e^*)(w^* - b)]^\eta e^*\bar{L}$. This is the Y that entered the demand curves (6) from the very beginning. Just as in a classic macro model or in the Solow growth model, output is in each period determined entirely from the supply side.

It is worth noting how much Fig. 11.6 looks like a traditional demand and supply diagram for the labour market. Here we have the rate of employment rather than employment itself along the horizontal axis, and the wage curve has taken the place of the labour supply curve, while the price-setting curve has taken the place of the labour demand curve.

There is a particular case of some interest where the wage curve in Fig. 11.6 becomes vertical so the rate of employment (and unemployment) is determined by the wage curve alone. If the rate of unemployment benefit is not fixed as a given amount in real terms, but as a given fraction of the real wage rate, $b = cw$, $0 \leq c < 1$, then the wage curve in (15) becomes: $w(1 - e - \eta) = (1 - e)cw$, giving:

$$e^* = \frac{1-c-\eta}{1-c} \text{ and hence } u^* = 1 - e^* = \frac{\eta}{1-c}. \quad (17)$$

We arrive at an explicit expression for u^* , and we see that the natural rate of unemployment is increasing in the replacement ratio, c , and in the wage elasticity of effort, η . Furthermore, the formula can be used to say something about orders of magnitude. For a (realistically) small value of η of 1 per cent, $\eta = 0.01$, and a replacement ratio of $\frac{2}{3}$, one gets $u^* = 0.03$, while, for instance, $\eta = 0.01$ and $c = 0.5$ gives $u^* = 2$ per cent, and 2–3 per cent is a substantial part of real-world structural unemployment which is typically in the range 5–7 per cent. In Exercise 2 below you will be asked to investigate further the realism of the efficiency wage model as an explanation of the natural rate of unemployment.

Main implications and structural policy

One of the main conclusions from the macroeconomic efficiency wage model is that *there is necessarily unemployment in equilibrium*: as mentioned above, $u^* > \eta$ and by assumption $\eta > 0$. In the particular case of a fixed replacement ratio one can see from (17) that $u^* > 0$. Unemployment is thus not just a possibility, as in the partial equilibrium model considered in Section 2. Unemployment is a necessary property of the macroeconomic equilibrium. Why is it that there *must* be unemployment in the macroeconomic equilibrium?

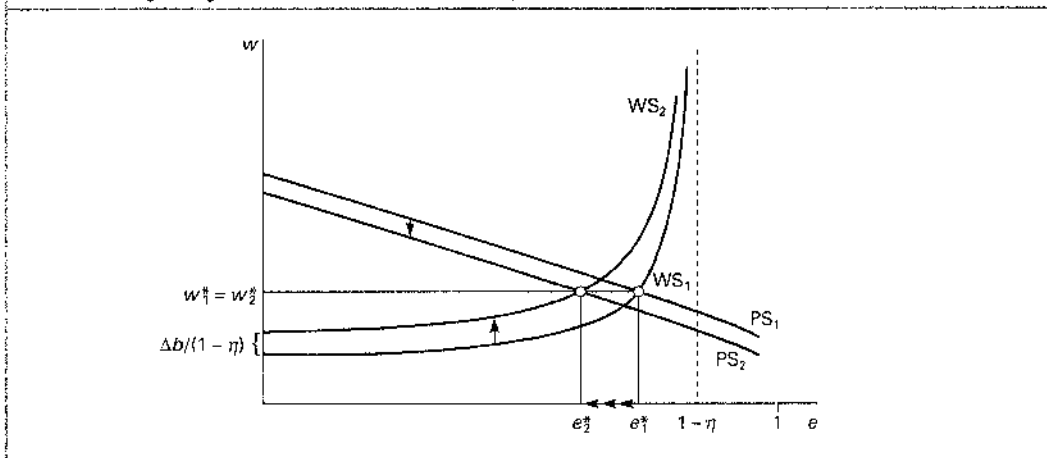
The key lies in the fact that workers only deliver a positive level of productivity if they are paid above the normal income, $v = ub + (1 - u)w$. Each individual firm therefore has an incentive to set its real wage w_i above v ; otherwise it will earn zero profits. If $u = 0$, then the normal income is given by the general wage, w , alone, $v = w$, so each individual firm wants to set its wage w_i above the general wage level w . Thus, whenever unemployment is zero there will be an upward pressure on wages. At a positive level of unemployment (created by firms driving up wages), it is possible for each firm's individual pay, w_i , to be above $v = ub + (1 - u)w$, even when the w entering in v is equal to the individual firm's wage, w_i , as it must be in equilibrium (this requires, of course, that $w > b$). Basically, employers bid wages up to a level that is so high that they do not want to buy all the labour supplied.

Other main conclusions come from comparative statics. We will consider changes in b and σ , which are naturally associated with labour market policy and competition policy, respectively. We will not consider changes in η , since it is difficult to imagine a kind of policy that can affect η .

Consider an increase in b . We have already noted that this shifts the wage curve WS upwards and the price-setting curve PS downwards, as illustrated in Fig. 11.7, where the wage and price-setting curves shift from WS_1 to WS_2 and from PS_1 to PS_2 , respectively.

In the new equilibrium there has to be a lower rate of employment, that is, a higher natural rate of unemployment. (You should go carefully through the economic mechanisms underlying this result.) It does not follow from the directions of the shifts of the curves whether the wage rate goes up or down. It so happens that the wage rate is just unchanged, a fact you will be asked to show in Exercise 5. The most important conclusion is, however, that according to the macroeconomic efficiency wage model *more generous unemployment benefits mean higher structural unemployment*.

Next consider an increase in σ , which means more elastic demand functions and therefore less market power for the firms, reflected in lower mark-ups, $m = \sigma/(\sigma - 1)$. We can think of the reduction in market power as caused by an intensified competition policy by the government. This shifts the price-setting curve PS upwards, but it does not affect the position of the wage curve. (In this case we will leave both the figure and the intuitive explanations to you.) In the new equilibrium there will be less unemployment and higher real wages. According to the macroeconomic efficiency wage model *stronger competition in product markets means lower structural unemployment and higher real wages*. It is worth emphasizing that conditions in the *product* markets, namely the degree of competition, influence the rate of structural unemployment, since structural unemployment is often viewed as a pure *labour* market phenomenon that can only be affected by labour market policies.

FIGURE 11.7 Higher b gives lower e^* in macroeconomic equilibrium

UNEMPLOYMENT AND STRUCTURAL POLICY IN THE GENERAL EQUILIBRIUM EFFICIENCY WAGE MODEL

- There is by necessity a strictly positive level of unemployment in the macroeconomic efficiency wage model. This is due to the positive dependence of worker productivity on the excess of pay over normal income and to the latter being a mix of average wages and income opportunities for the unemployed.
- According to the general equilibrium efficiency wage model, the rate of structural unemployment will be higher the more generous unemployment benefits are.
- According to the same model, more intense competition on product markets will imply lower structural unemployment as well as higher real wages.

11.5 Summary

1. Under the standard assumption that labour productivity does not depend on the wage rate, a profit-maximizing firm that can choose the wage it pays would never set the wage higher than strictly required for recruiting the workers it wants. Otherwise the wage could be lowered without reducing the labour supply to the firm, which would lower costs and raise profits. The firm would thus always position itself on the labour supply curve, creating full employment through its optimizing behaviour.
2. If labour productivity depends positively on pay, the above argument breaks down. Raising the wage above the level required to recruit the desired workforce has two counteracting effects on profits: the cost of labour increases, but at the same time productivity also increases. It may be that the second impact on profits is the strongest. The phenomenon that labour productivity (or effort) depends positively on the wage is referred to as efficiency wages. With a positive link from wages to productivity, it is a logical possibility that an individual firm will want to raise wages above the market-clearing level.

3. When labour productivity depends on real wages, each firm will set its real wage in accordance with the Solow condition saying that the elasticity of productivity with respect to the real wage should be equal to 1, and set employment at the optimal level given this wage rate. In a partial equilibrium model of the labour market, this behaviour could well imply a total level of employment smaller than the total labour supply induced by the efficiency wages set by firms. Under efficiency wages, unemployment in a full equilibrium with adjusted real wages is therefore a possibility, but in a partial labour market model unemployment is not a necessary feature of equilibrium.
4. In the partial labour market model analysed in the chapter, the equilibrium real wage rate is insensitive to supply and demand shocks to the revenue functions of firms, so employment bears the full burden of adjusting to such shocks. Efficiency wage models can therefore help to explain the fact that real wages fluctuate much less than employment over the business cycle.
5. There are at least four reasons why productivity should depend positively on pay. For the individual firm, setting a relatively high wage can be a way of (i) keeping good workers in the firm, thereby reducing recruiting and training costs, (ii) getting applications from relatively good workers in hiring situations, (iii) motivating workers to comply with a high level of required effort, because a larger difference between the wage paid and what can be earned outside the firm increases the cost to a worker of being fired for 'shirking', and (iv) motivating workers to exert a high level of effort for pure reasons of reciprocity.
6. Under plausible assumptions on the effort cost function, the 'shirking' model of efficiency wages implies that the efficiency function linking effort to pay should be an increasing, concave function of the excess of the real wage over the outside option, defined as the normal income a worker can expect to earn outside the firm where currently employed. The other motivating factors for the efficiency wage phenomenon point to similar curves.
7. The efficiency wage effect may be built into a general equilibrium macroeconomic model where the revenues of firms derive from consumer demand, and where the individual worker's outside option is a mixture of the general wage level and the unemployment benefit received in case of unemployment. In such a model unemployment is a necessary property of a macroeconomic equilibrium with fully adjusted real wages: if there were no unemployment, the outside option from the point of view of each firm would be the general wage level. To induce any effort at all, each firm would then have to set its own wage above the general wage level, but not all firms' wages can be above the average level. The upward pressure on wages would make labour so expensive that firms would not want to buy all the labour supplied. Unemployment would arise, and with positive unemployment it is possible for each firm to set a wage above the common outside option, since unemployment benefits are lower than wages.
8. The general equilibrium model with efficiency wages implies that more generous unemployment benefits will increase the natural rate of unemployment, whereas intensified competition in product markets will lower it. These features have implications for labour market and competition policies.

11.6 Exercises

Exercise 1. Plausible efficiency curves

Consider again Fig. 11.2 with an 'efficiency curve' and rays with slope $a(w)/w$. Can you draw a strictly increasing efficiency function that goes through $(0, 0)$, such that there is still a unique interior optimal wage rate, w^* ? Also draw one such that the optimal wage rate is zero. Finally draw one such that there is no optimal wage rate because the firm would always like to increase the wage rate further. Try to give reasons why the first of your curves may be reasonable, while the latter two are not.

Exercise 2. A partial/general equilibrium efficiency wage model

In this exercise an efficiency wage model with one firm and a given revenue curve is considered. This is like in a *partial equilibrium* model. However, one should think of the firm as *representative*, and therefore *general equilibrium* effects can be brought into the analysis.

The representative firm has a revenue function, $zR(aL)$, with notation and assumptions as in Section 11.2, and an efficiency function:

$$a = \left(\frac{w' - v}{v} \right)^\eta, \quad 0 \leq \eta < 1,$$

where v is the employees' outside option, and w' is the real wage paid by the representative firm. (This is for $w' \geq v$, while a is 0 for $w' < v$.)

1. Write down the first-order conditions for the firm's profit maximization problem with respect to w' and L . Derive the Solow condition in this context and give the expression for the optimal wage as a function of z , v , and η .

Assume that the outside option is $v = (1 - u)w + ucw$, $0 < c < 1$, where u is the unemployment rate, c is the replacement ratio, and w is the general real wage level.

2. Using how v relates to u , w and c , express the optimal wage rate of the representative firm as a function of η , u , w and c . Then use the 'equilibrium condition', that the general wage level entering in v must equal the wage set by the representative firm, to derive the equilibrium rate of unemployment, u^* . How do the parameters η and c affect u^* ? Explain the economic mechanisms involved.
3. What is the equilibrium productivity, a^* ? Assume that total labour supply is some fixed, inelastically supplied $\bar{L}$. By using your earlier findings, among them the relevant first-order condition from Question 1 above, find an expression for the equilibrium wage rate, w^* . (It is acceptable just to express w^* as a function of a^* , u^* and $\bar{L}$, but you can trace w^* back to parameters using your expressions for a^* and u^* .)
4. Assume that there is a trendwise movement upwards over time in the productivity factor z . Describe how this affects u^* and w^* over time. Why can your finding here be said to be a nice feature of the model?
5. Compute the equilibrium unemployment rate u^* for the following values of c and η :

$c \backslash \eta$	0.01	0.02
0.8		
0.5		
0.2		
0		

Comment on your results with respect to the statement: 'Even for realistically small values of η , and for plausible values of the replacement ratio, the efficiency wage model can explain a substantial part of real world natural unemployment'.

6. Insert the expression you found for u^* into $v = (1 - u)w + ucw$, and write w^* for the w on the right-hand side. This expresses the outside option as a function $v(w^*)$ of the equilibrium general wage level. By inserting $v(w^*)$ in place of v , write down the effort a in each firm as a function, $a(w', w^*)$, of the firm's own wage, w' , and the general wage level, w^* . How does the function look, e.g. for $\eta = 0.02$? To obtain any effort at all, how large a percentage of the general wage level, w^* , must the firm pay? Do you find this realistic?

Exercise 3. Deriving the demand side of the macro economy

Consider a representative consumer or household who has a utility function over the consumptions, $c_1, \dots, c_n$, of n different commodities:

$$C(c_1, \dots, c_n) = n \left(\frac{1}{n} \sum_{i=1}^n c_i^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \quad \sigma > 0, \sigma \neq 1. \quad (18)$$

1. Show that this utility function is homogeneous of degree one with respect to $c_1, \dots, c_n$.

Let the money prices, $P_1, \dots, P_n$, of the n commodities be given. Assume that the total amount of money that the representative consumer spends on the n commodities is M . Utility maximization must imply that the consumptions, $c_1, \dots, c_n$, are chosen to:

Maximize $C(c_1, \dots, c_n)$ subject to $P_1 c_1 + \dots + P_n c_n = M$.

Because of the homogeneity of C , one can solve this problem in two steps. First one can find the consumptions, $c_i^*(P_1, \dots, P_n)$ for $i = 1, \dots, n$ by which one unit of utility can be bought as cheaply as possible, that is, one solves the problem of choosing $c_1, \dots, c_n$ to:

$$\text{Minimize } P_1 c_1 + \dots + P_n c_n \text{ subject to } C(c_1, \dots, c_n) = 1, \quad (19)$$

and defines the price index $P(P_1, \dots, P_n)$ as the total cost of one unit of utility:

$$P = P(P_1, \dots, P_n) \equiv \sum_{i=1}^n P_i c_i^*(P_1, \dots, P_n).$$

Then one can use that from homogeneity, the amount of money M can buy exactly M/P units of utility, and that the optimal way to acquire each unit is by the consumptions $c_i^*(P_1, \dots, P_n)$, $i = 1, \dots, n$, so that the demand for commodity i must be:

$$c_i(P_1, \dots, P_n, M) = c_i^*(P_1, \dots, P_n) \frac{M}{P(P_1, \dots, P_n)}.$$

In everything that follows you can take as granted that the first-order conditions for the optimization in (19) are necessary and sufficient for a solution, $c_1^*, \dots, c_n^*$.

2. Write down the Lagrange function for the cost minimization problem in (19) using the definition of C in (18), and show that the first-order conditions consist of the constraint $C(c_1, \dots, c_n) = 1$, and for $i = 1, \dots, n$:

$$P_i = \lambda \frac{c_i^{\frac{1}{\sigma}}}{\sum_{j=1}^n c_j^{\frac{1}{\sigma}}}, \quad (20)$$

where λ is a Lagrange multiplier. (Hint: on the way you must use once more how C is given by (18), and that $C = 1$.)

3. Multiplying by c_i on both sides of (20), find $P_i c_i$, and add up over all commodities, to show that the minimal total cost of one unit of utility must fulfil:

$$P = \sum_{i=1}^n P_i c_i = \lambda.$$

4. In the first-order condition (20) above, write P for λ , and also use that from (18), $C(c_1, \dots, c_n) = 1 \Leftrightarrow \sum_{i=1}^n c_i^{(\sigma-1)/\sigma} = n^{1/\sigma}$, to show that:

$$c_i^*(P_1, \dots, P_n) = \left(\frac{P_i}{P} \right)^{-\sigma} \frac{1}{n}.$$

5. Use the result in Question 4, and the definition of the price index to show that:

$$P(P_1, \dots, P_n) = \left(\frac{1}{n} \sum_{i=1}^n P_i^{1-\sigma} \right)^{\frac{1}{1-\sigma}},$$

and that:

$$c_i(P_1, \dots, P_n, M) = \left(\frac{P_i}{P} \right)^{-\sigma} \frac{1}{n} \frac{M}{P},$$

where we have (again) just written P for $P(P_1, \dots, P_n)$.

(When all firms charge the same prices, $P_i/P = 1$ for all i , demand per commodity is $M/(nP)$, and hence total real demand is M/P . In a simple model where demand only comes from the households, this must in a symmetric equilibrium be the GDP. Writing Y for MP , we arrive at the demand functions (6) of the text.)

6. How does the ratio, c_i/c_j , between the consumptions of commodities i and j depend on the relative price, P_i/P_j , according to the consumer's demand? What is the elasticity of c_i/c_j with respect to the price ratio, P_i/P_j ? Why is the utility function in (18) called a constant elasticity of substitution utility function?

Exercise 4. How large are realistic values of the wage elasticity of productivity?

Given the efficiency function $e(w_i) = (w_i - v)^\eta$ for individual firms, and the simple form $v = ub + (1 - u)w$ for the normal income or outside option, we found in Section 11.4 the particular form for the wage curve:

$$w = \frac{u}{u - \eta} b,$$

where b is a fixed amount (in real terms). Explain how this wage curve was found. Find the elasticity of the real wage, w , with respect to the rate of unemployment u , as this depends on η and u . According to estimates, this elasticity is realistically around -0.1 , and rates of unemployment can be assumed to be more or less between 5 and 10 per cent. What is the implied interval for η , the elasticity of productivity with respect to real wage?

Exercise 5. Solving for the macroeconomic equilibrium

In Section 11.4, and in Exercise 2, it was shown how one can find an explicit expression for the equilibrium rate of unemployment, u^* , when the replacement ratio, c , is fixed. In the

case where the benefit rate, b , is fixed one can also find an explicit expression for the equilibrium rate of unemployment, u^* , it is just a bit more cumbersome. The equilibrium rate of employment, e^* , and the equilibrium real wage, w^* , are the solutions in e and w to the two equations (15), the wage curve, and (16), the price-setting curve. One can solve directly for u and w using (14) and (16). Show that this gives:

$$u^* = \frac{\frac{1}{\eta^{1-\eta}}}{\frac{1}{\eta^{1-\eta}} - bm^{\frac{1}{1-\eta}}} \quad \text{and} \quad w^* = \frac{1}{\eta} \left(\frac{\eta}{m} \right)^{\frac{1}{1-\eta}}.$$

It is assumed that the parameters are such that the denominator in u^* is positive. This is really an important restriction (making the whole model meaningful) that we have not mentioned in the chapter. Show that the restriction is equivalent to $b < w^*$. The parameter restriction is thus equivalent to the condition underlying a positive mark-up of wage over benefit in equilibrium, so it is the parameter restriction underlying Footnote 8.

How do changes in b and m (coming from σ) affect u^* and w^* ? Explain intuitively. What is the expression for u^* if you insert $b = cw^*$?

Exercise 6. Systematic productivity changes in the macroeconomic efficiency wage model

We said in this chapter that for the efficiency wage model to be in accordance with the stylized facts of growth, a systematic (trendwise) change in productivity should build itself fully into the real wage. We also said that an appropriately formulated general equilibrium macro model of efficiency wages would have this feature, but when we formulated our macroeconomic efficiency wage model in Section 11.4 we did not consider changes in productivity. This exercise considers the model of Section 11.4 with a productivity shift parameter included.

The model we will consider is the same as the model in Section 11.4 in all but three respects. The first change is that the production function in each sector i is changed into:

$$Y_i = za_i L_i,$$

where z is a productivity shift parameter common to all sectors. We interpret z as a systematic and predictable parameter, changing for instance permanently from one level to another, or increasing by a certain constant rate over time. Our second revision is that the efficiency (or effort) function of each firm i makes productivity depend on the *relative* excess of the real wage over the outside option:

$$a_i = \left(\frac{w_i - v}{v} \right)^\eta$$

Finally we assume that the benefit rate b changes proportionally with productivity:

$$b = \bar{b}z, \quad \bar{b} > 0,$$

reflecting that empirically benefit levels do tend to evolve in line with general welfare in the long run.

1. Argue that in the presence of systematic productivity changes (e.g. a trendwise increase over time) it should, from the motivations for efficiency wages, be the *relative* excess of wage over outside option that governs effort.

2. Set up the profit maximization problem of firm i and show that its optimal (real) price and wage are now:

$$P_i = m \frac{w_i}{z \left(\frac{w_i - v}{v} \right)^\eta},$$

$$w_i = \frac{v}{1 - \eta},$$

where m has the same definition as in Section 11.4. Compare with the parallel expressions from the model of Section 11.4. Explain the differences.

3. Assume, as in Section 11.4, that the outside option for each sector is: $v = ub + (1 - u)w$, where w is the average wage level. Show that this leads to the wage curve:

$$w = \frac{u}{u - \eta} \bar{b}z.$$

Show that the price curve is:

$$mw = z \left(\frac{u(w - \bar{b}z)}{w - u(w - \bar{b}z)} \right)^\eta,$$

where we assume that $w > \bar{b}z$. Compare again with the parallel expressions in Section 11.4.

4. Combining the wage curve and the price curve show that the macroeconomic equilibrium is:

$$w^* = \frac{z}{m} \left(\frac{\eta}{1 - \eta} \right)^\eta$$

$$u^* = \frac{\eta}{1 - \frac{\bar{b}m}{\left(\frac{\eta}{1 - \eta} \right)^\eta}}$$

Write down a parameter restriction ensuring our assumption $w^* > \bar{b}z$, and show that this also implies $u^* > \eta$. Comment on the macroeconomic equilibrium with respect to how the real wage and structural unemployment are affected by systematic productivity shifts. Given a total labour supply of $\bar{L}$, what is the GDP (Y) in the macroeconomic equilibrium and how is GDP affected by z ?

5. Explain that the entire analysis in this exercise extends to the case, and therefore can be interpreted to cover the case, of a given replacement ratio where $b = cw$, if one sets the replacement ratio c as:

$$c = \frac{\bar{b}m}{\left(\frac{\eta}{1 - \eta} \right)^\eta}.$$

How are real wages and unemployment affected by changes in the replacement ratio? Write the natural rate of unemployment as a function of η and c . What is the condition for $u^* < 1$? Is this a realistic condition?

Exercise 7. 'Unemployment as a worker discipline device'

Carl Shapiro and Joseph E. Stiglitz gave their seminal paper on the shirking model of efficiency wages the title 'Equilibrium Unemployment as a Worker Discipline Device' (*American Economic Review*, 74, 1984). Try to give an explanation of this use of language by going through (once more) the explanation why there has to be unemployment in equilibrium in the macroeconomic model of Section 11.4, and think of the motivation for the efficiency function, $a(w_i) = (w_i - v)^\eta$, as coming from the shirking model explained in Section 11.3.

Exercise 8. Taxes in the macroeconomic efficiency wage model

This exercise introduces an income tax in the macroeconomic efficiency wage model. The proportional income tax rate is τ , so the efficiency function of firm i is:

$$a(w_i) = ((1 - \tau)w_i - v)^\eta, \quad 0 < \eta < 1.$$

Firms face demand curves as in (6), so the real after-tax profits of firm i are:

$$\Pi_i = (1 - \tau)p_i^{-\sigma} \frac{Y}{n} \left(p_i - \frac{w_i}{((1 - \tau)w_i - v)^\eta} \right).$$

The task of firm i is to choose p_i and w_i to maximize real profits, taking τ , Y , n and v as given.

1. Derive the firm's optimal wage and interpret the result. The real rate of unemployment benefit before tax is b , and b is assumed to be a constant fraction of the general wage level w , $b = cw$, $0 < c < 1$. If unemployment benefits are not taxed, the outside option is:

$$v = ub + (1 - u)(1 - \tau)w,$$

and we assume in this case that $1 - \tau > c$.

2. Derive the rate of structural unemployment, u^* , in the macroeconomic equilibrium. Explain how u^* is affected by the taxation and give the intuition.
3. Now assume that unemployment benefits are taxed by the same rate as other income. What is the value of the outside option, v ? Find the natural rate of unemployment and explain how and why the result differs from what you found in Question 2.